USN | | | | | | | | | | 23CSE204

SECOND SEMESTER B.Tech EXAMINATION, JUNE-2024

DIGITAL LOGIC DESIGN

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 x02=10**		
1.	**a.**	What is Number System? Discuss the importance of number systems.	L2	CO1	**02**
	b.	Draw the Logic circuit for the expression using Basic gates. F= $X\bar{Y} + \bar{X}Z$	L2	CO2	**02**
	c.	Write the Sum and Carry expressions for the Half Adder.	L2	CO3	**02**
	d.	Write a logic circuit for NOR gate, SR Latch.	L2	CO4	**02**
	e.	Define Sequence Recognizer. What are the advantages of it.	L2	CO6	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Convert the following Gray Numbers into Binary Numbers: **i.** (1011) **ii.** (101110)	L2	CO1	**05**
		OR			
	b.	Perform Binary Subtraction of following using 2's complement method: **i.** $(1010.010)_2$ - $(111.111)_2$ **ii.** $(11000)_2$ - $(10001)_2$	L2	CO1	**05**
3.	**a.**	State and Prove De-Morgan's Theorem.	L2	CO2	**05**
		OR			
	b.	With a Logic symbol, Truth table, and Logical expression explain the Basic Logic gates.	L2	CO2	**05**
4.	**a.**	What are combinational circuits? Differentiate between Combinational and Sequential Circuits.	L3	CO2	**05**
		OR			
	b.	Explain working of Full Subtractor with suitable logic circuit and truth table.	L3	CO3	**05**
5.	**a.**	Design a priority encoder for a system with 2 inputs.	L3	CO3	**05**
		OR			

Deemed-to-be University Learning Resource Centre Jain Global Campus Ramanagara

	b.	Explain the working of Octal to Binary Encoder with a neat Logic circuit.	L2	CO3	**05**
6.	**a.**	Differentiate between Synchronous and Asynchronous Counters.	L3	CO5	**05**
		OR			
	b.	Write a short note on the Applications of Shift Registers.	L2	CO4	**05**
7.	**a.**	Define sequential machine? Explain types of sequential machine	L2	CO6	**05**
		OR			
	b.	Design a 2 bit Synchronous binary Counter	L4	CO6	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Convert the following: **i.** $(1101.10)_2 = (?)_{10}$ **ii.** $(95.4)_{10} = (?)_2$ **iii.** $(ABCD0)_{16} = (?)_2$ **iv.** $(156.7)_8 = (?)_2$ **v.** $(101011)_2 = (?)_8$	L2	CO1	**10**
		OR			
	b.	Perform using 2' complement: $(9)_{10}$ - $(4)_{10}$, $(27)_{10}$ - $(75)_{10}$	L2	CO1	**10**
9.	**a.**	Convert the Boolean function F (A, B, C) = Σ (2,4,5,6) into Product of Sum form. Minimize the following function using the K-map. f (A, B, C, D) =Σ m (0,3,4,5,7) + d (8,9,10,11,12,13,14,15)	L2	CO2	**10**
		OR			
	b.	What are universal gates? Implement XOR and XNOR gates using universal gates.	L2	CO2	**10**
10.	**a.**	What is MUX? Design 4 to 1 MUX using logic gates. Write the truth table and explain it's working.	L3	CO3	**10**
		OR			
	b.	What is a Full Adder? Design a full adder with two half Adders.	L3	CO3	**10**
11.	**a.**	With a schematic, discuss the operation of Bi-directional shift Register.	L3	CO4	**10**
		OR			

	b.	With a logic diagram explain the concept of Universal shift register.	L3	CO4	**10**
12.	**a.**	Explain the working principle of Decade counter with a neat logic diagram.	L3	CO5	**10**
		OR			
	b.	Design a synchronous counter with a sequence 4,7,3,0,2 using J K Flip flops.	L3	CO5	**10**
13.	**a.**	Design a synchronous counter using D flip flops that goes through states 0, 1, 2, 4, 0. The unused states must always go to zero on the next clock pulse.	L4	CO6	**10**
		OR			
	b.	Design a non-overlapping 101 sequence detector	L4	CO6	**10**